

Lab 2 — TA Prep Sheet

CS460 Summer 2026 · MySQL Setup & Python Connector · Likely student questions and quick answers

How this lab runs: students read the lab page and install everything themselves. Your opening line: *"Have a look at Lab 2 and follow the steps. If you hit something, the Troubleshooting section probably has it. If you've checked there and you're still stuck, just ask. Once your test query and Python script both work, you're free to go."*

This sheet is the questions they'll actually walk up and ask. Skim it once before the lab so you can answer in one breath.

The single most common one, say it up front to the whole room: "If you set a MySQL password during install, open `test_mysql.py` and put it between the quotes on the `password=' '` line. If you left it blank, leave it as is." This preempts half the Python failures.

Install & version questions

Q Which file do I download? There are like six options.

A Mac: the ARM64 DMG if it's an M1/M2/M3/M4, x86 if it's an older Intel Mac. They can check under Apple menu → About This Mac → "Chip". Windows: the MySQL Installer, choose Developer Default. That's it, ignore the rest.

Q Do I have to make an Oracle account to download?

A No. On the download page scroll to the bottom and click "No thanks, just start my download".

Q My MySQL is version 9.x but the page / Workbench mentions 8.4. Is that a problem?

A No. 9.x is fine for everything in this course. They're fully compatible for the SQL we use.

Q Workbench shows a yellow warning that my server version is incompatible.

A Click OK and ignore it. Workbench was last built for MySQL 8.0, your server is newer, but only advanced migration tools are affected and we never touch those. This is in Troubleshooting under "version warning".

Q Should I use the DMG or Homebrew on Mac?

A Either. Homebrew is faster if they already have it (`brew install mysql`). The DMG is fine too. Homebrew installs tend to have a blank root password, which is okay for this lab.

Password questions

Q It's asking for a root password and I don't remember setting one.

A If they installed via Homebrew, it's probably blank, tell them to just press Enter / leave it empty. If they installed via DMG or Windows installer, they set one during install. If they truly can't recall it, the cleanest fix is reinstall and set a memorable one.

Q When I type my password nothing shows up. Is it broken?

A That's normal in the terminal, passwords are hidden as you type, no dots or stars. Just type it and press Enter. (In Workbench's password box you do see dots, this only applies to the terminal.)

Q I get "ERROR 1045: Access denied for user root".

A Wrong password. Have them retype carefully. If genuinely forgotten, reinstall is the fastest path for a lab setting.

Connection questions

Q Workbench says "Can't connect to MySQL server".

A The server isn't running. Start it: Mac Homebrew `brew services start mysql`; Mac DMG via System Settings → MySQL; Windows `net start MySQL`. It's in Troubleshooting under "Can't connect".

Q There's no "Local instance" tile on the Workbench home screen.

A Click the "+" next to MySQL Connections and make one: Host `127.0.0.1`, Port `3306`, Username `root`. Test connection, then save.

Q "command not found: mysql" in the terminal.

A MySQL is installed but not on their PATH. This only matters if they use the terminal, Workbench works regardless. Troubleshooting has the per-OS PATH fix. Key reminder: they must open a NEW terminal after changing PATH.

SQL / database questions

Q Where do I type the SQL?

A The middle panel of Workbench, the SQL editor. Paste a block, click the lightning-bolt button to run it. Results show at the bottom.

Q My CREATE TABLE failed saying the table already exists.

A They ran it twice. Either ignore it (the table's already there) or drop it first with `DROP TABLE missions, destinations, spaceships;` and re-run. Order matters because of foreign keys, drop

missions first.

Q The foreign key ALTER fails.

A Usually they ran it twice (the key already exists) or the referenced table doesn't exist yet. Make sure all three CREATE TABLEs ran successfully first.

Q My query returns nothing / empty.

A They probably skipped the INSERT step, or didn't run it. No rows in, no rows out. Have them re-run section 4d, then the test query.

Q I don't see space_missions in the sidebar.

A Click the refresh icon next to SCHEMAS in the Navigator. Then double-click `space_missions` so it goes bold (that makes it the active database).

Python questions (where most real problems are)

Q "error: externally-managed-environment" when I run pip install.

A Their Python blocks system-wide installs to protect itself. This is exactly why the lab uses a virtual environment. Make sure their prompt shows `(cs460env)` before running pip. If not, re-run the activate line. Troubleshooting covers this.

Q "ModuleNotFoundError: No module named 'mysql'" but I installed it.

A They installed with one Python and ran with another. Fix: activate the venv, reinstall inside it, run the script from inside it. All three from the same activated environment. This is the single biggest Python gotcha, expect it a lot.

Q Python says it can't find my file, but the file is right there on my Desktop.

A TextEdit secretly saved it as `test_mysql.py.rtf`. Tell them to recreate it with the `cat` command in step 5c, or in TextEdit do Format → Make Plain Text before saving. Classic Mac trap.

Q What is the virtual environment even for? Why can't I just pip install?

A It's an isolated Python sandbox so libraries install cleanly without breaking their system Python. Modern Python and Homebrew require it. One sentence: "It keeps this project's packages separate so pip works and nothing else breaks."

Q Do I have to activate the environment every time?

A Yes, every new terminal window. The `(cs460env)` prefix in the prompt tells them they're inside it. If it's missing, re-run the activate command.

Q What's the point of the Python connector? We already used Workbench.

A Workbench is for them typing queries by hand. The connector lets their Python programs query MySQL automatically. Every lab after this uses it, so we verify it works today rather than the night

before homework.

"Am I done?" questions

Q How do I know I'm finished?

A Point them at the "You're done when..." checklist at the bottom of the page. Four things: `mysql --version` works, Workbench connects, the test query returns Explorer/Jupiter/Research, and `python test_mysql.py` prints the tuple. All four green, they can leave.

Q Do I need to submit anything?

A [Confirm with the instructor, but typically:] nothing to submit for a setup lab, or a screenshot of the working test query / Python output. Set this expectation at the start.

The lab page's Troubleshooting section now covers 22 specific errors grouped by step (install, Workbench, connecting, Python, general), each with a copy-paste fix. Most students will self-resolve from there. If someone is stuck for more than a couple of minutes on something genuinely off-script, it's usually a corporate/locked-down laptop blocking the install, fall back to Docker or a partner's machine and have them finish at home before Lab 3.